Experiment 7

Random Sampling and Sampling Distribution

```python
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Generate a population (e.g., normal distribution)
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_size)

# Step 2: Random sampling
sample_sizes = [30, 50, 100]  # different sample sizes to consider
num_samples = 1000  # number of samples for each sample size

sample_means = {}

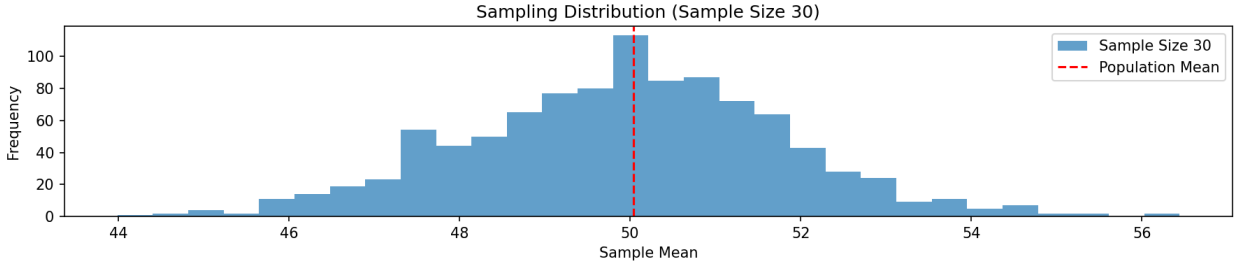
for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size, replace=False)
        sample_means[size].append(np.mean(sample))

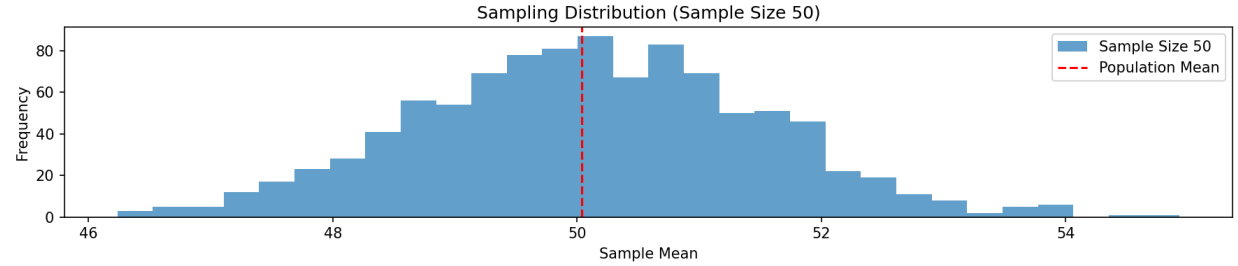
# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))

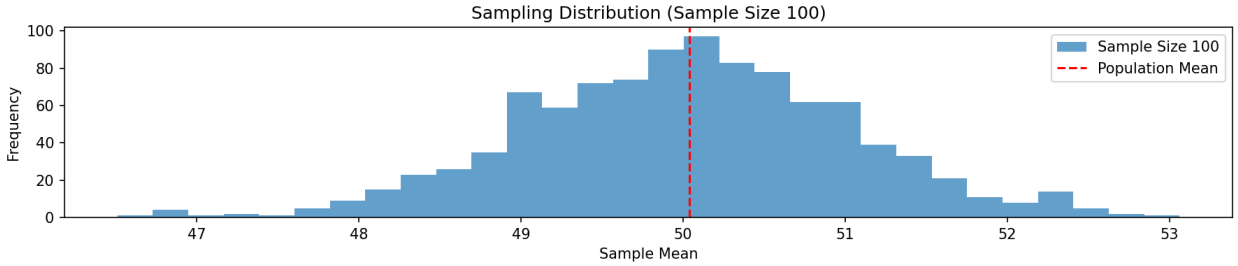
for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i+1)
    plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
    plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5, label='
    Population Mean')
    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

plt.tight_layout()
plt.show()
```

7.2s


Sampling Distribution (Sample Size 30)
Sample Size 30
Population Mean
Frequency
Sample Mean


Sampling Distribution (Sample Size 50)
Sample Size 50
Population Mean
Frequency
Sample Mean


Sampling Distribution (Sample Size 100)
Sample Size 100
Population Mean
Frequency
Sample Mean